Control loop troubleshooting guide

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1. Purpose

This guide helps operators, instrument technicians and control engineers find out why a basic control loop performs badly and what to check in the field. It follows the order an engineer should work in. First confirm the measurement, then the final element (valve and actuator), then the tuning, and only then the process design. Most poor loops are caused by the valve or the measurement, not by the tuning.

2. First checks for any loop

1. Is the loop in automatic? A loop in manual (MODE = MAN on the OPC UA server) cannot control. Ask the shift why it is in manual before changing anything (section 4.4).
2. Is the controller output (OP) at a limit (0 % or 100 %) for long periods? Then the valve has run out of range and no tuning can help (section 4.3).
3. Does the process variable (PV) follow the setpoint (SP) within the expected band? Compare with the normal setpoint in the control narratives (DOC-CN-001).
4. Was anything changed recently? Look at the management of change log, the maintenance history (DOC-MH-001) and the shift log.
5. Plot PV, SP and OP together over at least 4 hours. Most patterns below are recognized on this one plot.

3. Reading the PV, SP and OP plot

Pattern on the plot	Most likely cause	Section
PV and OP cycle continuously with a regular period, also when the SP is constant	Aggressive tuning, or valve stiction	4.2, 4.1
OP ramps slowly while PV stays flat, then PV jumps (saw-tooth OP, square-wave PV)	Valve stiction (static friction)	4.1
OP stays at 100 % (or 0 %) and PV is below (or above) SP	Valve or actuator saturation, undersized valve	4.3
MODE is MAN for hours, OP is flat	Operators do not trust the loop	4.4
PV has a fast fuzzy band, OP moves constantly with small steps	Measurement noise	4.5
PV stays at SP but the real	Sensor drift	4.6

Pattern on the plot	Most likely cause	Section
process moves away slowly (seen in lab results or a second instrument)		
Most of the variance comes right after SP changes	Setpoint activity, not a control fault	4.7
PV moves away after an upstream change and comes back slowly	Load disturbance	4.8
Two loops on the same vessel move together; tuning one upsets the other	Loop interaction	4.9

4. Causes, symptoms and field checks

4.1 Valve stiction (static friction)

Symptoms. The OP ramps while the PV does not move, then the valve breaks free and the PV jumps past the setpoint. The loop then cycles with a saw-tooth OP and a square-wave PV. The cycle continues with a constant setpoint. The period is set by how long the integral action takes to build enough force to move the stem.

Why it happens. Tight or worn packing, especially graphite packing, deposits on the stem or in the trim, an undersized actuator, or a positioner that is badly calibrated. Graphite packing gives good emission performance but more friction than PTFE V-ring packing.

Loops with known stiction in this plant: FIC-1003 (recycle flow, valve FV-1003), FIC-1102 (reflux flow, valve FV-1102), FIC-1206 (quench flow, valve FV-1206) and FIC-1214 (side draw flow, valve FV-1214). All four valves have graphite packing (DOC-DS-001).

Field checks. 1. Put the loop in manual with operations' agreement. Move the OP in steps of 0.5 % and record when the stem or the flow moves. A dead band of 2 % or more confirms stiction. 2. Read the positioner diagnostics (HART): friction estimate, travel deviation, supply pressure. 3. Check the packing nuts: overtightened packing is the most common cause after maintenance. 4. Inspect the stem for scratches or deposits at the next opportunity.

What to do. Adjust the packing or calibrate the positioner. As a temporary measure, reduce the integral gain so the cycle is slower and smaller. Retuning alone does not remove stiction. Do not raise the controller gain: it makes the cycle faster, not smaller.

4.2 Oscillation from aggressive tuning

Symptoms. PV and OP oscillate with a regular, smooth (sinusoidal) shape. The amplitude grows after a setpoint change and then settles into a steady cycle. Unlike stiction, the OP is not a saw-tooth.

Loops with known oscillation: PIC-1002 (separator V-1004 pressure), PIC-1103 (compressor suction pressure), PIC-1202 (reactor inlet pressure) and TIC-1220 (cooling water return

temperature). PIC-1002, PIC-1103 and PIC-1202 had their controller gain raised in the 2025 turnaround to speed recovery after upsets (DOC-CN-001, Rev. B). The cycling started after that change. TIC-1220 cycles because of interaction with the cooling water consumers.

Field checks. 1. Put the loop in manual for 10 minutes. If the cycle stops, the cause is inside the loop (tuning or valve). If it continues, it comes from outside (another loop or the process). 2. If the cycle stops in manual, do the stiction test of section 4.1 to rule out the valve.

What to do. Reduce the controller gain by about 30 % and repeat a setpoint step test. Aim for a response with less than 10 % overshoot. For pressure loops with long dead time (PIC-1002 has about 28 s), keep the integral action slow.

4.3 Valve saturation (output at the limit)

Symptoms. The OP stays at or near 100 % (or 0 %) and the PV cannot reach the setpoint. The integral action winds up, and when the load drops the PV overshoots.

Loops with known saturation: LIC-1004 (separator V-1004 level), LIC-1113 (reflux drum level), FIC-1109 (steam flow), PIC-1207 (compressor discharge pressure) and FIC-1219 (cooling water flow). These valves were sized for the original capacity. The capacity increase project raised the flows beyond their comfortable range.

Field checks. 1. Compare the OP with the real valve position (positioner feedback, local indicator). A valve that does not open fully at 100 % OP may have a travel stop, a supply pressure problem or a damaged actuator. 2. Check upstream conditions. A blocked strainer or low pump discharge pressure reduces the flow the valve can pass. 3. Check whether a manual bypass is partly open or closed against the design.

What to do. If the valve opens fully and still cannot pass the flow, the valve is undersized: plan a larger trim or a larger valve (higher Cv). Until then, check that anti-windup is enabled in the DCS so recovery is fast.

4.4 Loop kept in manual

Symptoms. MODE stays MAN for long periods. The OP is flat or moved in steps by the operator. The variable is controlled by hand.

Loops often in manual: TIC-1005 (reactor R-1005 temperature), PIC-1110 (network pressure), PIC-1212 (flare header pressure) and LIC-1223 (equalization tank level). Operators switch TIC-1005 to manual during feed changes, after an event in which the loop overshot (incident report INC-2025-031).

What to do. Ask the shift supervisors why. Fix the cause they name (often an old oscillation or a bad measurement). Retune with a step test and agree an automatic-mode target with operations, for example 95 % of the time in automatic.

4.5 Measurement noise

Symptoms. A fast, fuzzy band on the PV. The OP follows the noise with many small moves, which wears the actuator and packing.

Loops with noisy measurements: AIC-1006 (product composition), AIC-1108 (stream composition), AIC-1205 (hydrogen purity) and AIC-1215 (sulfur content in product). These are analyzer loops. The noise comes from the sample conditioning system (pressure and flow at the sample point, filter fouling, bubbles).

Field checks. Check the sample flow and pressure against the analyzer datasheet, clean or change the sample filter, check the analyzer's validation log.

What to do. Fix the sample system first. Then set a first-order PV filter of 5 to 10 seconds in the DCS. Reduce the controller gain if the OP still moves too much.

4.6 Sensor drift

Symptoms. The PV agrees with the setpoint, but a second instrument, a lab result or the energy balance shows the true value moving away slowly. Drift is invisible on the PV plot alone.

Loops with known drift: PIC-1008 (header pressure), TIC-1111 (product temperature), TIC-1209 (furnace outlet temperature, thermocouple TT-1209) and AIC-1224 (treated effluent pH, probe AT-1224). Thermocouples in furnace service drift after long exposure at high temperature. pH probes drift as the reference junction ages.

What to do. Compare with a portable instrument or a redundant one. Recalibrate or replace the element. These instruments are on a 6-month calibration interval instead of 12 (DOC-DS-001, sheet Instruments).

4.7 Setpoint activity

Symptoms. The PV variance is high, but most of it comes right after setpoint changes. Between changes the loop holds well.

Loops with frequent setpoint changes: FIC-1007 (charge flow), FIC-1112 (production flow), FIC-1210 (fuel gas flow) and FIC-1222 (effluent flow to treatment). Their setpoints follow operating campaigns and production targets.

What to do. Judge the loop on its response to each change: rise time, overshoot, settling time. Do not retune because of the variance alone.

4.8 Load disturbances

Symptoms. After an upstream change (feed rate, feed quality, fuel gas heating value, steam consumers) the PV moves away and returns slowly.

Loops with recurring disturbances: TIC-1009 (exchanger outlet temperature), TIC-1104 (furnace temperature), LIC-1105 (surge drum level), TIC-1203 (catalyst bed temperature), LIC-1211 (flash drum level), PIC-1217 (medium-pressure steam pressure) and TIC-1226 (Claus reactor temperature).

What to do. Add feedforward from the measured disturbance where one exists, for example the charge flow FIC-1201 for TIC-1203. Check that the integral action is not too slow. Surge drum and flash drum levels should use averaging level control (wide band, slow tuning), not tight control.

4.9 Loop interaction

Symptoms. Two loops on the same equipment move together. Tuning one tighter makes the other worse.

Loops with interaction: LIC-1010 (interface level, V-1004), LIC-1107 (interface level in treating), LIC-1208 (compressor suction drum level), LIC-1218 (condensate drum level) and LIC-1225 (slop tank level). The OPC UA server publishes the interacting stream as `CPM.Loops.<loop>.SECONDARY`.

What to do. Tune the two loops at clearly different speeds: the faster one at least three times faster than the slower. Where that is not possible, add a decoupling term or consider a multivariable controller.

5. When to call whom

Finding	Who
Valve, positioner, packing	Instrument maintenance (work order)
Transmitter or analyzer calibration	Instrument maintenance, analyzer technician
Tuning	Process control engineer on duty
Valve undersized, process change	Process engineering, management of change
Loop kept in manual	Shift supervisor and process control engineer together